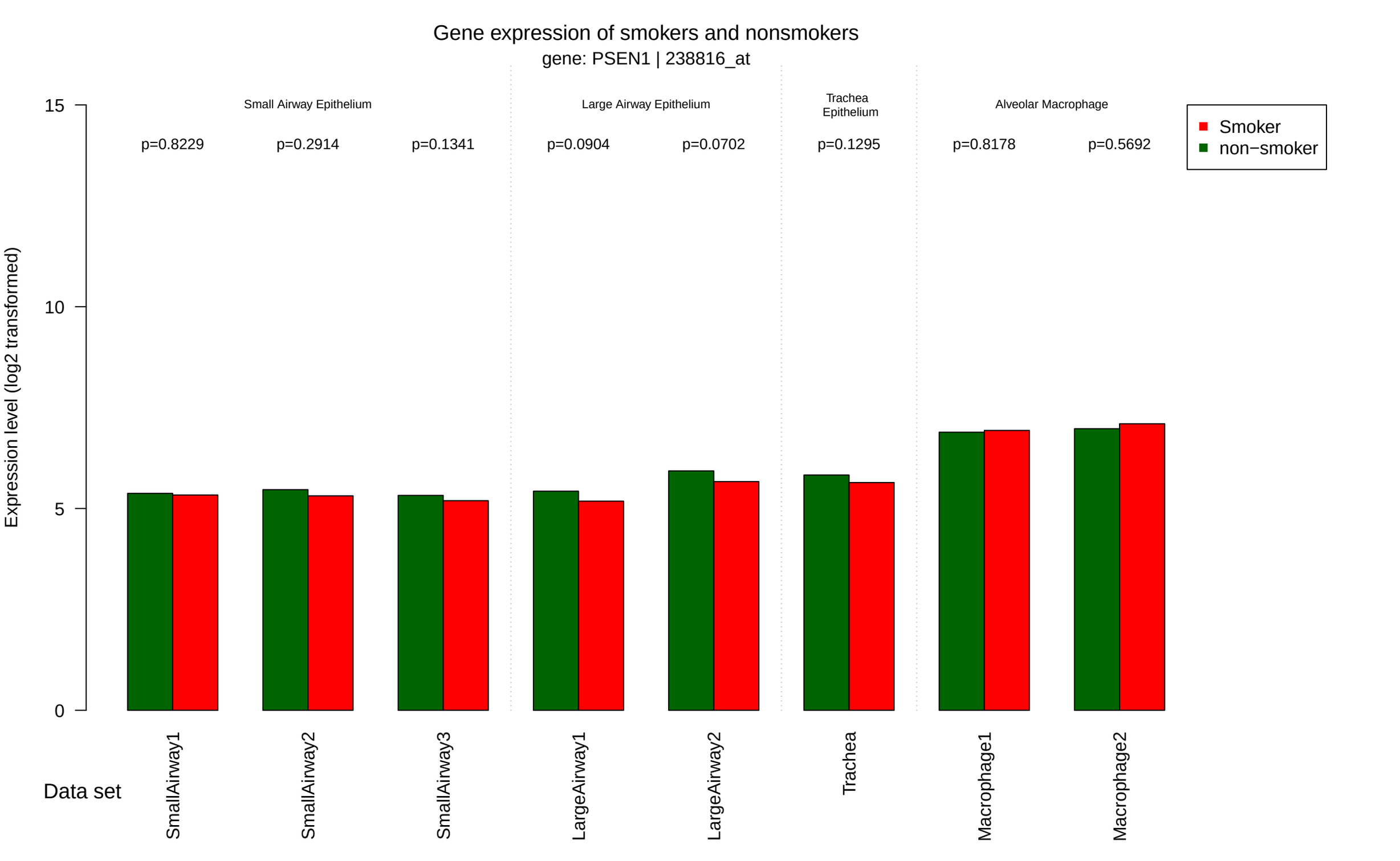
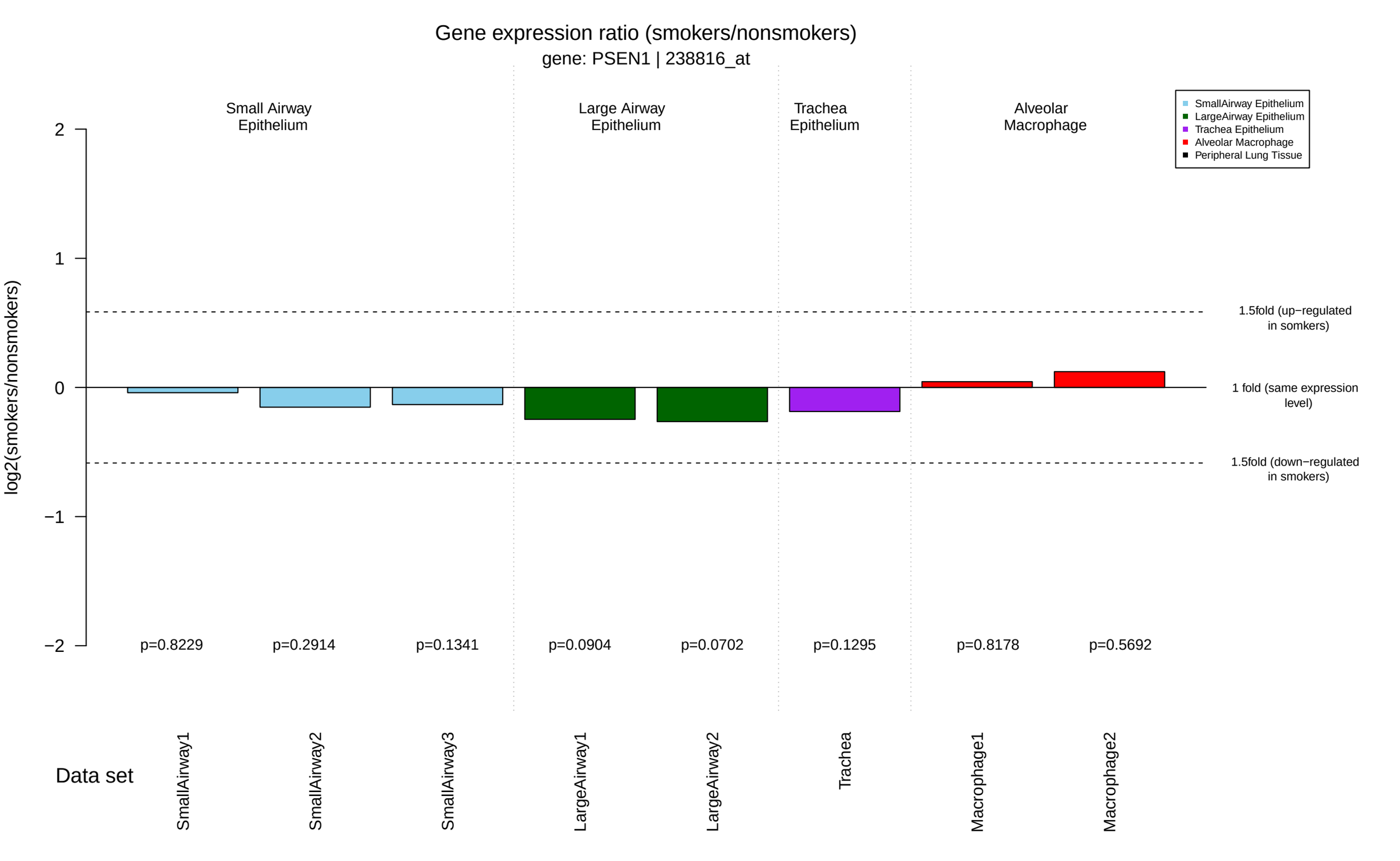
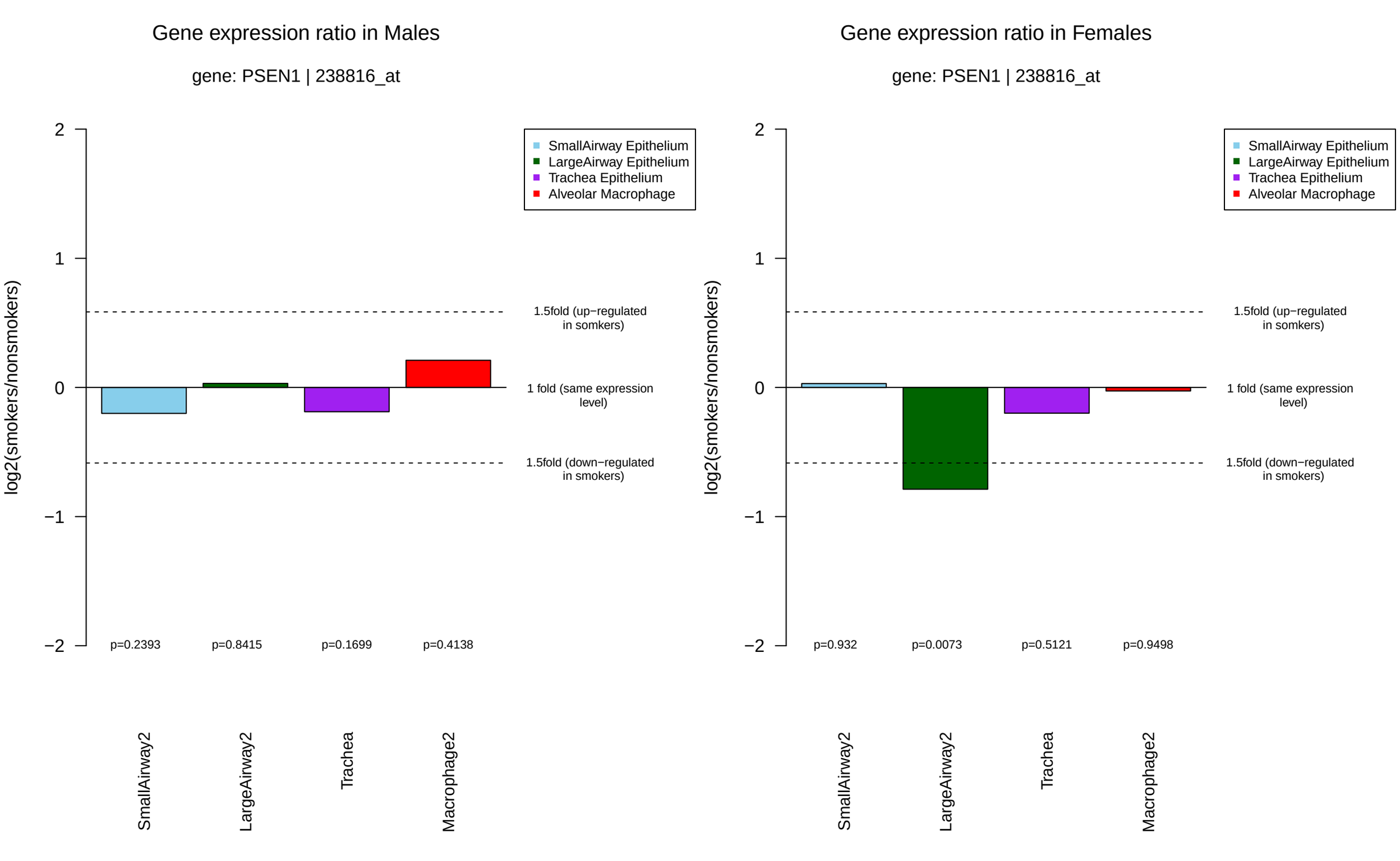
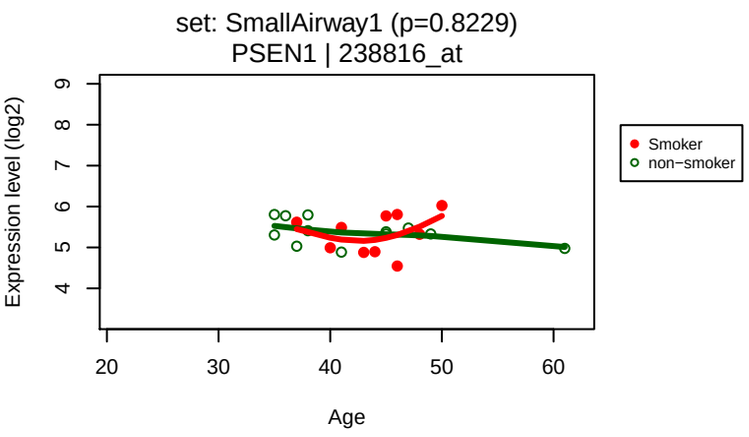
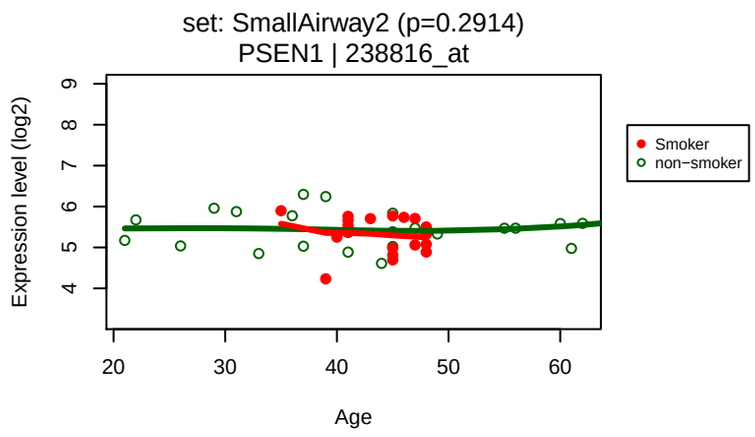
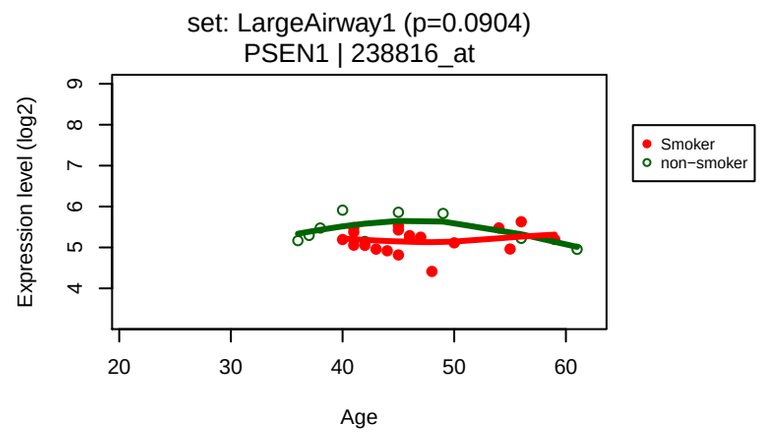

Gene expression of smokers and nonsmokers
gene: PSEN1 | 238816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8229
p=0.2914
p=0.1341
p=0.0904
p=0.0702
p=0.1295
p=0.8178
p=0.5692
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

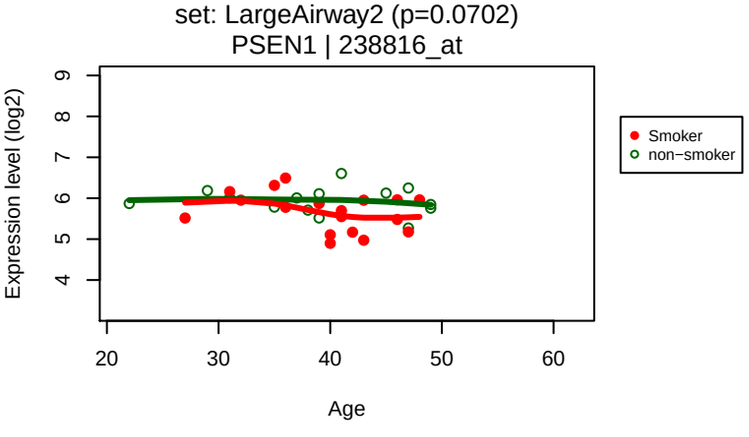

Gene expression ratio (smokers/nonsmokers)
gene: PSEN1 | 238816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8229
p=0.2914
p=0.1341
p=0.0904
p=0.0702
p=0.1295
p=0.8178
p=0.5692
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

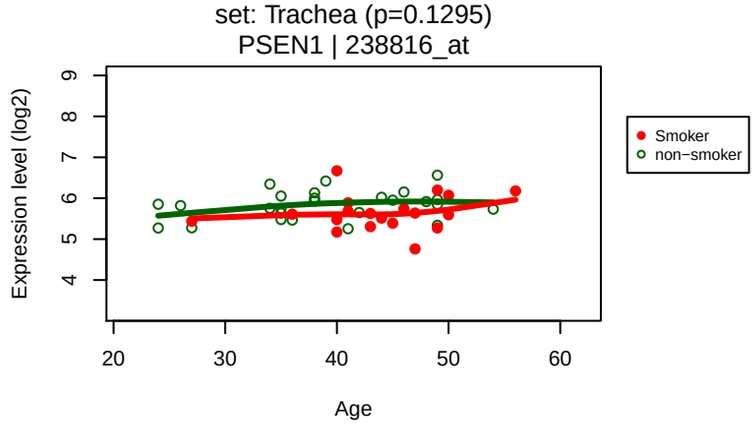

Gene expression ratio in Males
gene: PSEN1 | 238816_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2393
p=0.8415
p=0.1699
p=0.4138
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN1 | 238816_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.932
p=0.0073
p=0.5121
p=0.9498
SmallAirway2
LargeAirway2
Trachea
Macrophage2

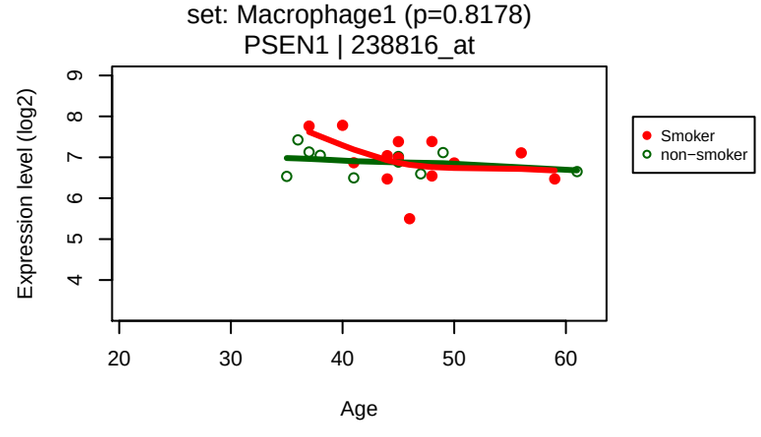

set: SmallAirway1 (p=0.8229)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker

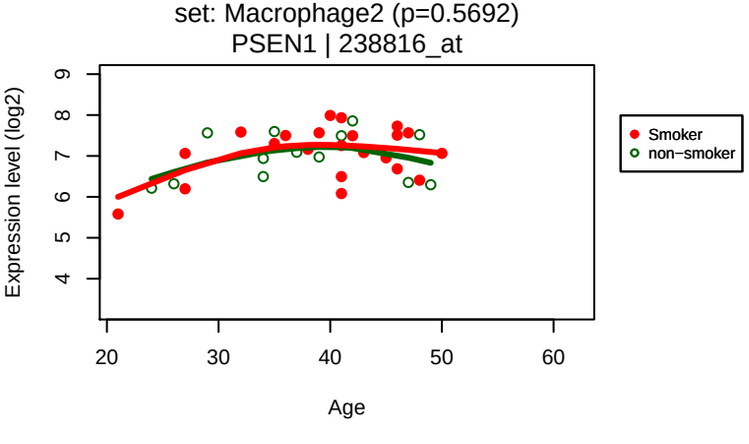

set: SmallAirway2 (p=0.2914)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker

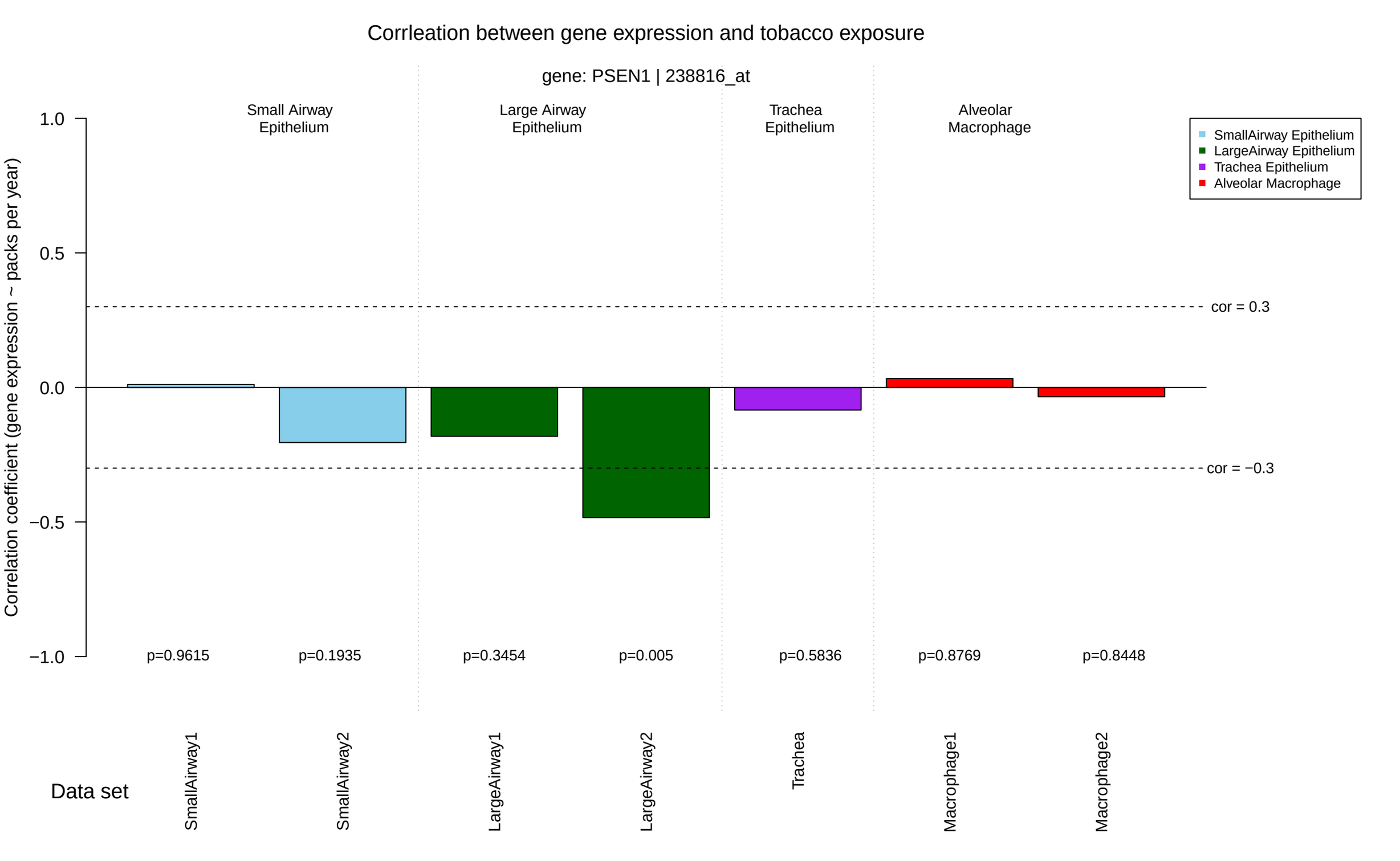

set: LargeAirway1 (p=0.0904)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0702)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1295)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8178)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5692)
PSEN1 | 238816_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN1 | 238816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9615
p=0.1935
p=0.3454
p=0.005
p=0.5836
p=0.8769
p=0.8448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2